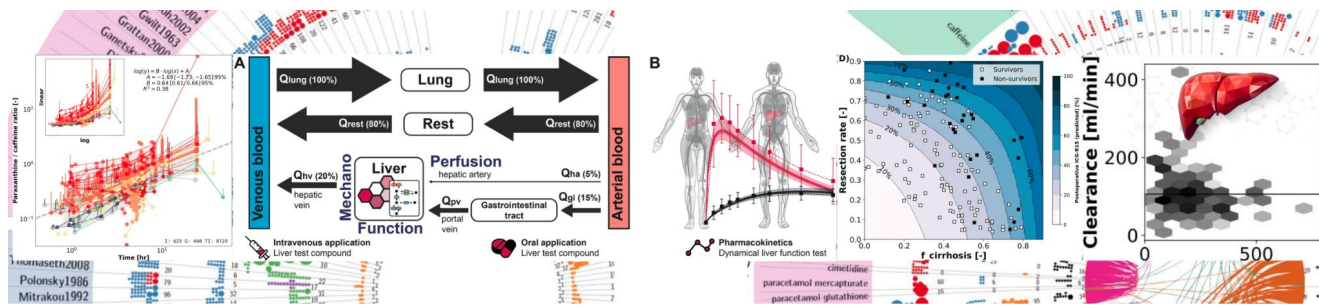




Pharmacokinetics Modeling Course

Absorption and Dosing



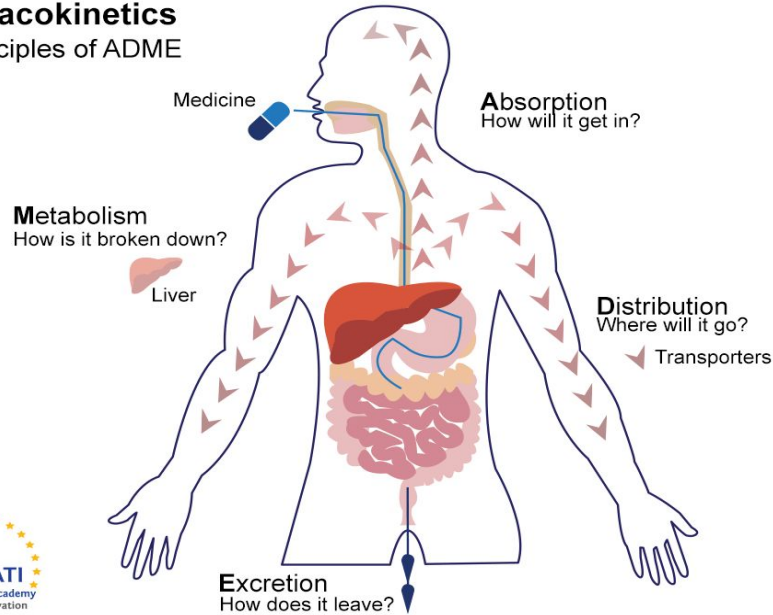
Dr. Matthias König
 Humboldt-University Berlin
 Systems Medicine of the Liver
 koenigmx@hu-berlin.de
<https://livermetabolism.com>

By the end of this section, you should be able to:

1. Understand the principles of **drug absorption** and factors that influence it.
2. Describe common **routes of administration** and their impact on absorption.
3. Explain the concept of **bioavailability** and **first-pass metabolism**.
4. Explore models of **oral absorption** (e.g., **first-order**, **zero-order**, **lag time**).
5. Understand the relationship between **dosing**, **absorption rate**, and **plasma concentration**.
6. Compare **single vs. multiple dosing** and their effects on drug levels.
7. Apply absorption and dosing concepts to **optimize drug therapy**.

Pharmacokinetics

The principles of ADME



ADME processes determine pharmacokinetics

- **Absorption**
- **Distribution**
- **Metabolization**
- **Elimination**

Administration Sites

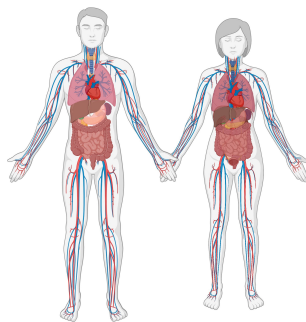
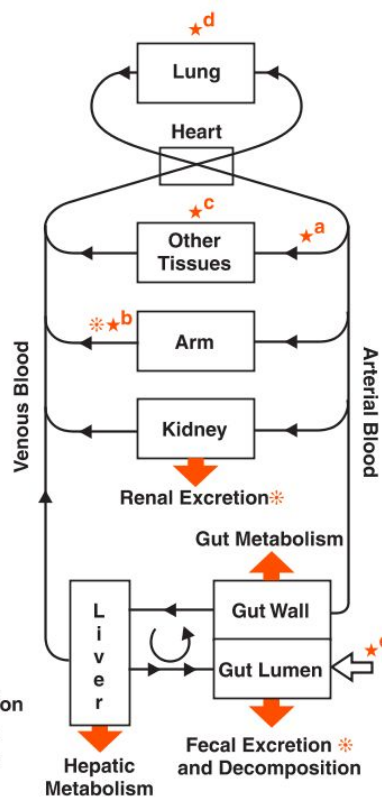


FIGURE 2-2 Once absorbed from any of the many sites of administration, drug is conveyed by blood to all sites within the body, including the eliminating organs. Sites of administration include: *a*, artery; *b*, peripheral vein; *c*, muscle and subcutaneous tissue; *d*, lung; and *e*, gastrointestinal tract, the most common route (denoted by *open arrow*). When given intravenously into an arm vein, the opposite arm should then be used for blood sampling. The movement of virtually any drug can be followed from site of administration to site of elimination.

many sites of administration

- a) artery
- b) peripheral vein (iv)
- c) muscle & subcutaneous tissue
- d) lung
- e) gastrointestinal tract (oral)

Application Forms: Contraceptives

Birth Control Pills

Birth control pills are the most widely used contraceptive method among women of all ages.

TYPES

Birth control pills come in **two types**, depending on the hormones they contain:

- **Combination pills:** estrogen and progestin
- **The minipill:** only progestin

HOW THEY WORK

The hormones in birth control pills prevent pregnancy by:

- Suppressing ovulation
- Thickening cervical mucus
- Thinning uterine lining

PROS

- 91% effectiveness with typical use
- Come in various types
- Do not hinder fertility
- Help regulate periods & relieve PMS

CONS

- Don't protect against STDs
- Easy to skip pills
- Require prescription
- May increase blood clot risk

www.shecares.com

Birth Control Patches

While commonly favored for their **convenience** and **effectiveness**, birth control patches may not be a suitable birth control method for all women.

HOW TO USE THEM

They are placed on the body on a 4-week schedule:

- **First 3 weeks:** new patch is put on the same day every week
- **4th week:** no patch is put on, which triggers a period

HOW THEY WORK

Contain estrogen and progestin, which are absorbed through the skin and into the bloodstream to:

- Prevent ovulation
- Stop the sperm from fertilizing the egg

PROS

- 91% effectiveness with typical use
- Only 3 applications per month
- Do not hinder fertility
- May improve irregular periods and acne

CONS

- Require diligent applications
- No protection from STDs
- May increase the risk of blood clots.
- May not be used by women with certain health conditions and other factors

www.shecares.com

Birth Control Rings

Birth control rings are small, flexible devices worn in the vagina to offer **safe and convenient protection** from unplanned pregnancy.

INSERTION

Vaginal rings are **placed in the vagina** by a woman (similar to tampons).

They're used on the following a 4-week schedule:

- Wear the ring for the **first 3 weeks**
- Remove the ring for the **first 3 weeks**

HOW DO THEY WORK

Contain **estrogen and progestin**, through the vaginal tissue to:

- Stop the ovary from releasing the egg
- Prevent sperm from fertilizing the egg

PROS

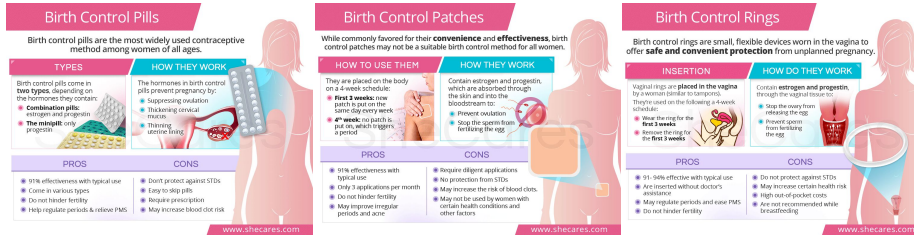
- 91- 94% effective with typical use
- Are inserted without doctor's assistance
- May regulate periods and ease PMS
- Do not hinder fertility

CONS

- Do not protect against STDs
- May increase certain health risk
- High out-of-pocket costs
- Are not recommended while breastfeeding

www.shecares.com

Application Forms: Contraceptives



Ethinylestradiol: synthetic estrogen used in combination contraceptives

Mechanism of Action: inhibits ovulation by suppressing the hypothalamic-pituitary-ovarian axis.

Various routes of administration

- **Oral:** most common route, typically in combination with a progestin in oral contraceptive pills.
- **Transdermal:** patches that release ethinylestradiol and a progestin through the skin.
- **Vaginal:** vaginal rings that release ethinylestradiol and a progestin directly into the vaginal tissues and bloodstream.
- **Injectable:** Occasionally used in combination with a progestin in long-acting injectable contraceptives

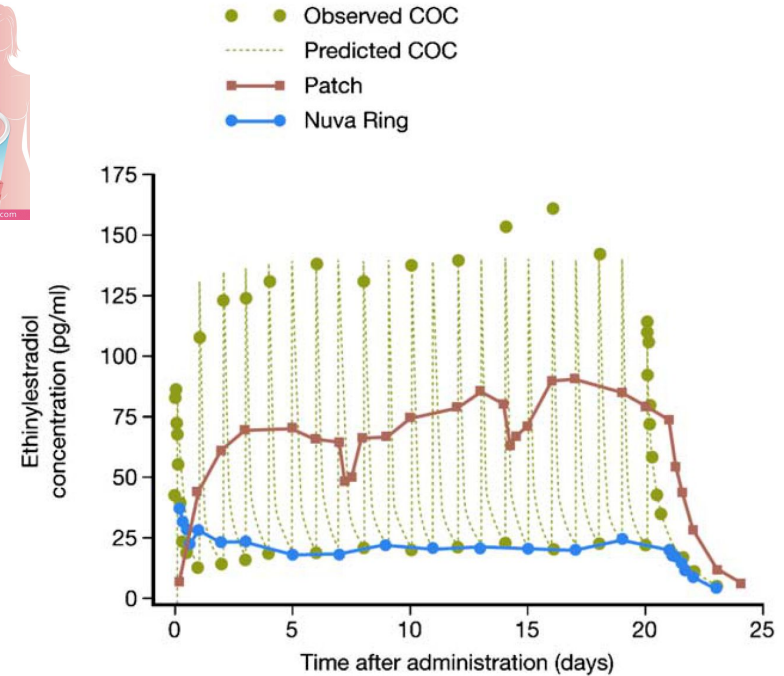


Fig. 2. Mean EE C-t curves for subjects (ASPE group) treated with NuvaRing ($n=8$), the transdermal contraceptive patch ($n=6$) and the COC ($n=8$).

van den Heuvel MW, van Bragt AJ, Alnabawy AK, Kaptein MC. **Comparison of ethinylestradiol pharmacokinetics in three hormonal contraceptive formulations: the vaginal ring, the transdermal patch and an oral contraceptive.** Contraception. 2005 Sep;72(3):168-74. doi: 10.1016/j.contraception.2005.03.005. PMID: 16102549.

Absorption

Various factors can influence drug absorption, including:

- 1. Route of Administration:** The route of administration significantly impacts how a drug is absorbed. For instance, drugs administered intravenously bypass the absorption process as they are introduced directly into the bloodstream. However, orally administered drugs must pass through the stomach and intestines, where they are absorbed into the bloodstream. This can be influenced by factors such as pH levels, presence of food, and gastrointestinal motility.
- 2. Drug Formulation:** The physical and chemical properties of the drug can influence how well it is absorbed. For example, drugs formulated in a liquid solution are often absorbed more rapidly than those in a tablet or capsule.
- 3. Physiological Factors:** Individual characteristics like age, sex, genetic factors, and health status can also influence drug absorption. For example, certain conditions like malabsorption syndromes or diseases affecting the liver or kidney can alter the absorption of drugs.
- 4. Drug Interactions:** Certain drugs can interact in the body and affect absorption. For instance, some medications can increase stomach acidity, which can affect the absorption of other drugs.

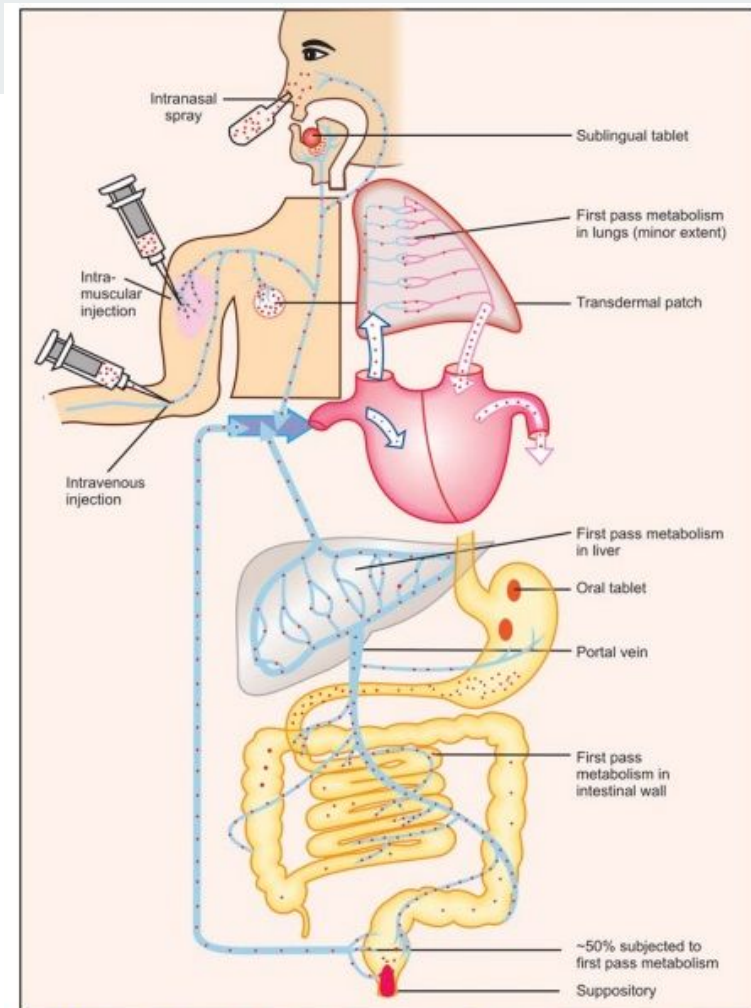


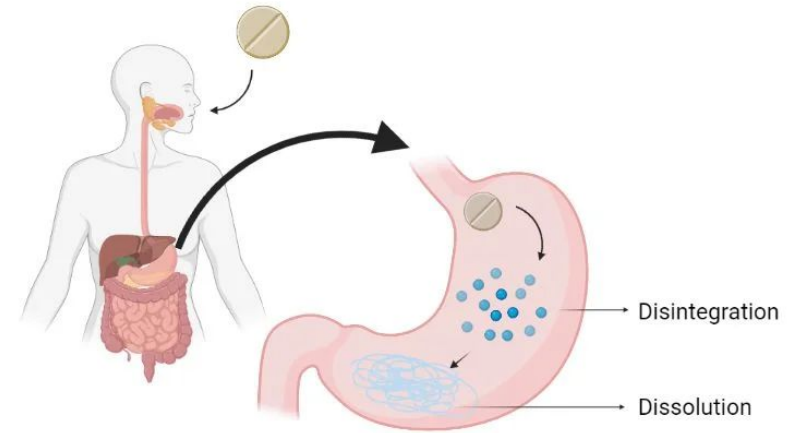
Fig. 1.1: Vascular pathway of drugs absorbed from various systemic routes of administration and sites of first pass metabolism

Dissolution of Oral Medication

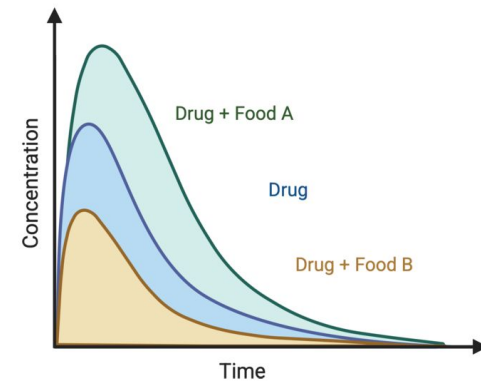
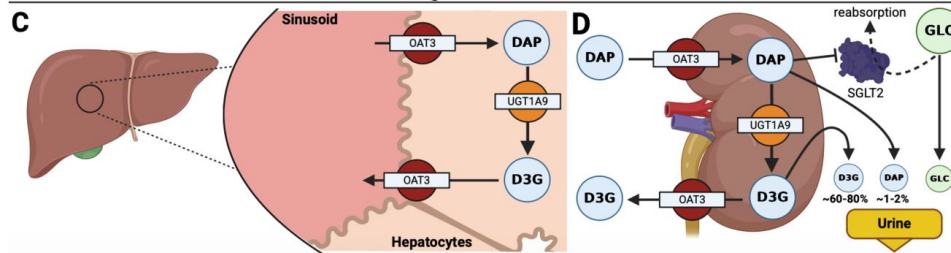
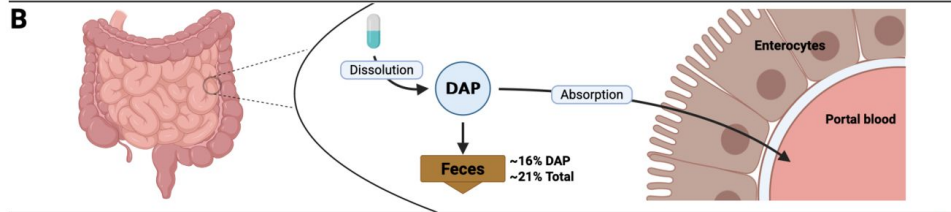
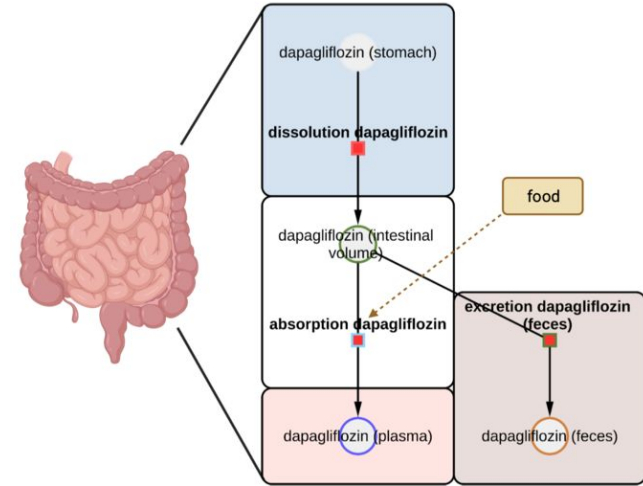
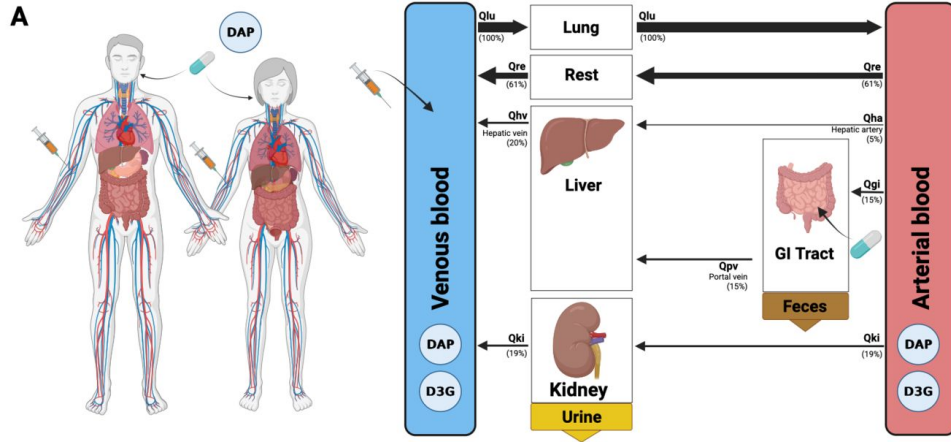
Dissolution: process by which a solid drug disintegrates and dissolves in a solvent (like gastrointestinal fluids) to form a solution that can be absorbed into the body.

- Dissolution is crucial for drug absorption in solid dosage forms.
- The dissolution rate depends on the drug's solubility.
- Drug **formulation** can significantly impact dissolution rate.
- The rate of dissolution can influence **drug bioavailability**.
- In vitro dissolution testing is used to estimate how a drug will dissolve and be absorbed in the body.
- Physiological conditions like pH, motility, and the presence of food can affect drug dissolution.

In essence, dissolution is vital in pharmaceutical science to ensure the drug dissolves properly for effective absorption and therapeutic effect.



Example: Food Effect Dapagliflozin



Example: Food Effect Dapagliflozin

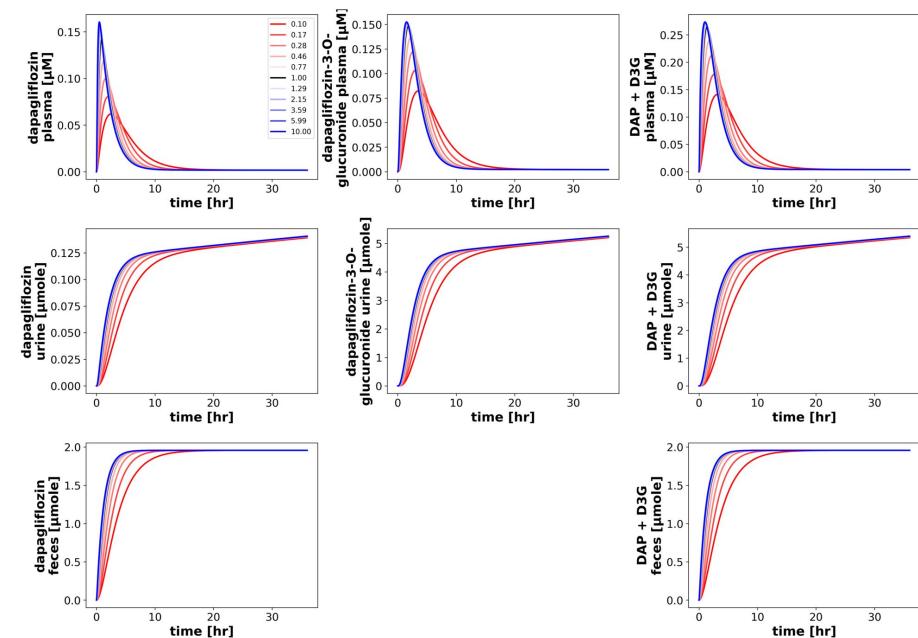


Figure 25: Pharmacokinetic simulations of dapagliflozin under food-effect conditions. Simulated concentrations of dapagliflozin (DAP), dapagliflozin-3-O-glucuronide (D3G) and dapagliflozin total (DAP+D3G) in plasma, urine and feces over time.

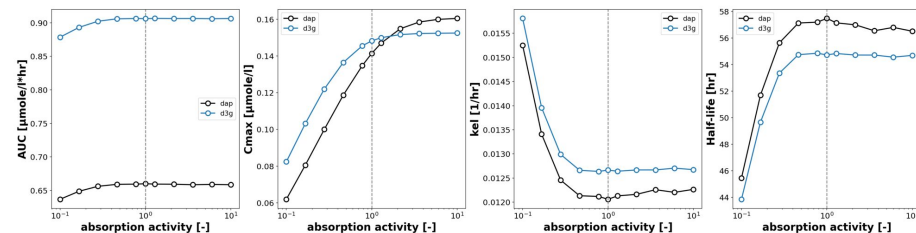
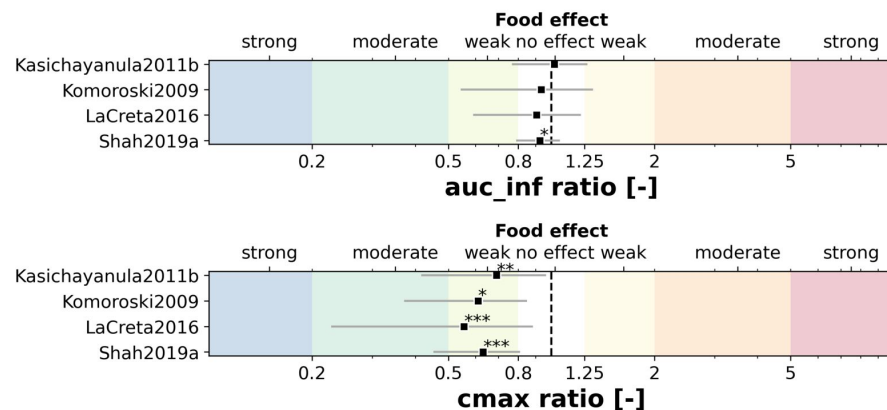
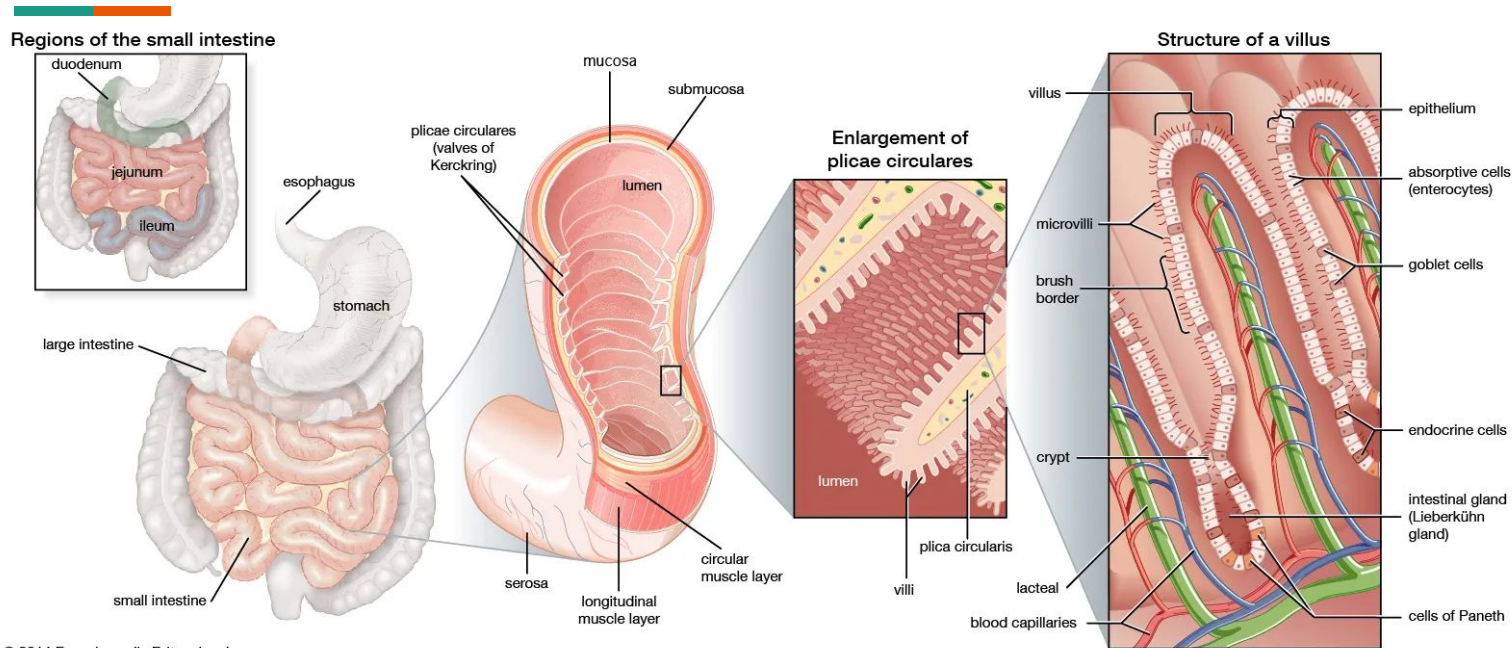


Figure 27: Pharmacokinetic parameters of dapagliflozin and dapagliflozin-3-O-glucuronide for absorption activity scan. AUC , C_{max} , kel and half-life for absorption activity of dapagliflozin (black) and dapagliflozin-3-O-glucuronide (blue).



Absorption of Oral Medication in the Intestine



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<https://www.pharmacistopinions.com/adme-in-pharmacokinetics/>
<https://www.britannica.com/science/small-intestine>

Site-dependent Absorption

- transporters/processes specific to certain intestinal segments
- E.g. ABCB1

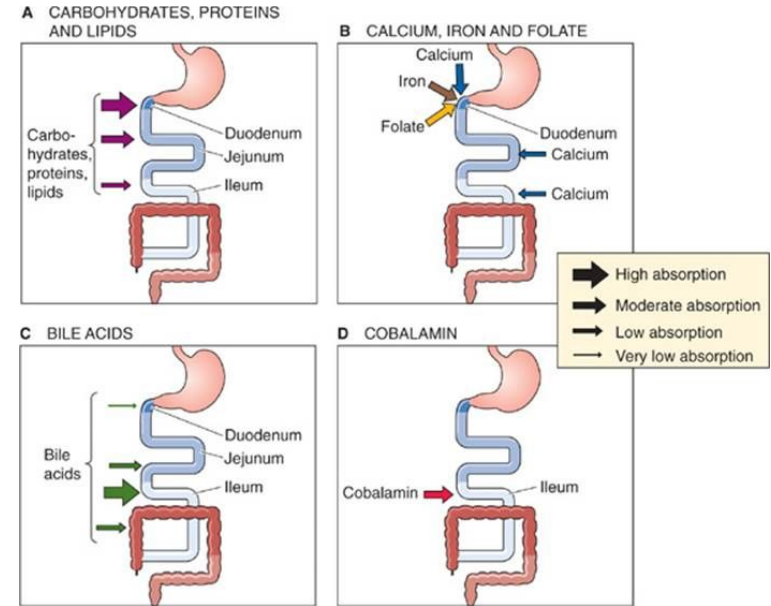
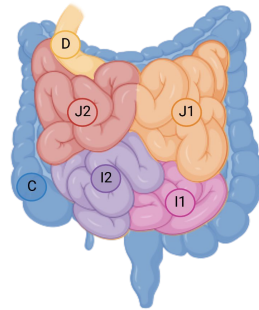
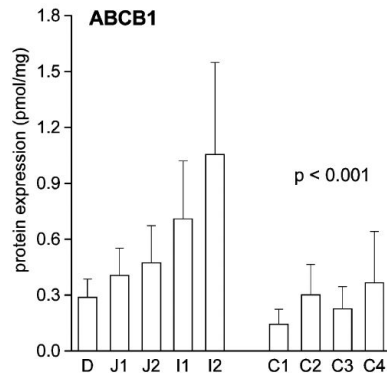


Figure 45-2 Sites of nutrient absorption. **A**, The entire small intestine absorbs carbohydrates, proteins, and lipids. However, the absorption is greatest in the duodenum, somewhat less in the jejunum, and much less in the ileum. The thickness of the *arrows in the inset* indicates the relative magnitude of total absorption at the indicated site in vivo. The maximal absorptive capacity of a specific segment under *optimized* experimental conditions (e.g., substrate concentrations) may be greater. **B**, Some substances are actively absorbed only in the duodenum. **C**, Bile acids are absorbed along the entire small intestine, but active absorption occurs only in the ileum. **D**, The vitamin cobalamin is absorbed only in the ileum.

Bioavailability

Bioavailability

Due to losses during absorption, intestinal metabolism, efflux and hepatic extraction only a fraction of the drug appears in the systemic circulation.

Clinical implications

- **Delays or loss of drug during absorption** can introduce a large variability in drug response.
- **Disease conditions** and **co-medications** may profoundly affect the absorption of certain drugs.

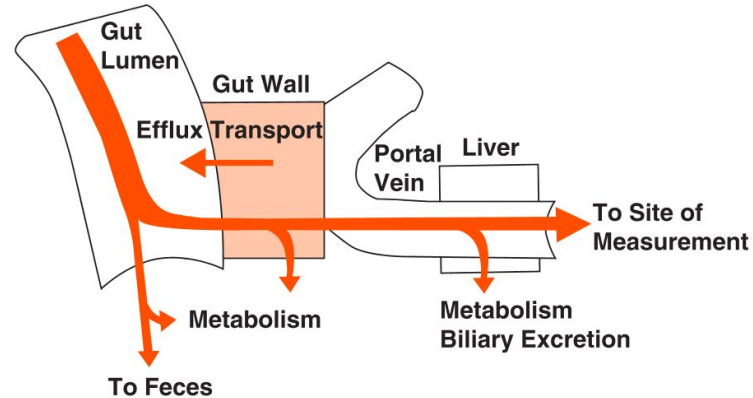


FIGURE 2-3 A drug, given as a solid, encounters several barriers and sites of loss in its sequential movement (colored arrows) through the gastrointestinal tissues and the liver. Incomplete dissolution, degradation in the gut lumen, metabolism by enzymes, and efflux by transporters, in the gut wall are causes of incomplete input into the systemic circulation. Removal of drug as it first passes through the liver may further reduce systemic input.

Example: oral semaglutide

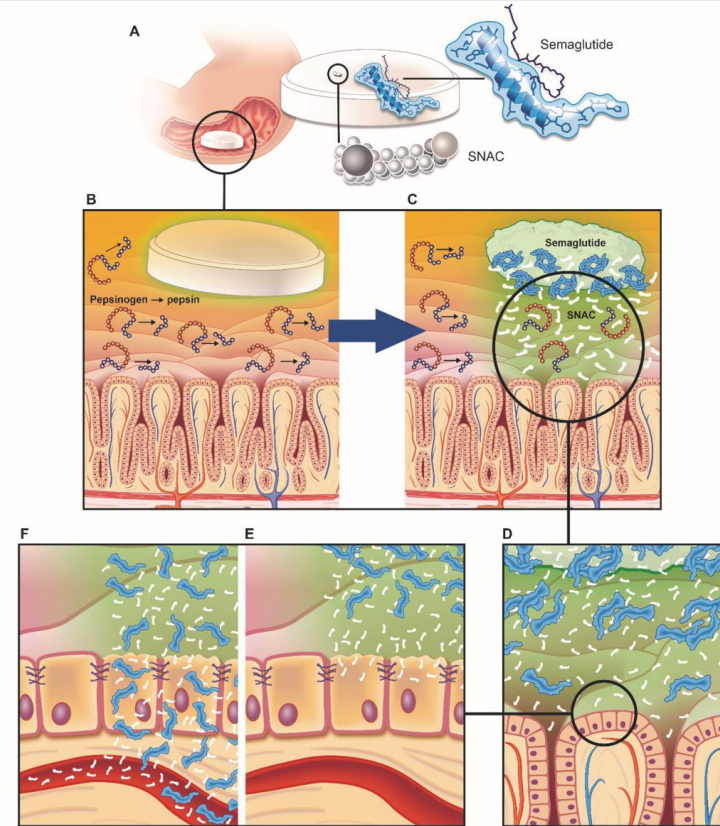
- first available oral GLP-1 agonist
- mimic the action of GLP-1, a hormone that enhances glucose-dependent insulin secretion, inhibits glucagon release, and slows gastric emptying

A) semaglutide is co-formulated with absorption enhancer SNAC

B-C) highly localized release of SNAC which neutralizes pH and prevents conversion of pepsinogen $\rightarrow$ pepsin

D) SNAC weakens semaglutide self-interactions

E-F) SNAC inserts in the mucosa membrane



Kim HS, Jung CH. **Oral Semaglutide, the First Ingestible Glucagon-Like Peptide-1 Receptor Agonist: Could It Be a Magic Bullet for Type 2 Diabetes?** Int J Mol Sci. 2021 Sep 14;22(18):9936. doi: 10.3390/ijms22189936.

Unerwartete Schwangerschaften unter Abnehmspritze

Montag, 3. Juni 2024



Newsletter abonnieren

Zur Startseite



/AntonioDiaz, stock.adobe.com

Berlin – Derzeit erregen national wie international Meldungen in den sozialen Medien Aufmerksamkeit, wonach Frauen unter dem Abnehm- und Diabetesmedikament Semaglutid unerwartet und ungeplant schwanger geworden seien. Das könne selbst unter Kontrazeptionsschutz geschehen, heißt es in Artikeln mit Schlagzeilen wie „Warum Ozempic-Babies‘ kein Grund zur Freude sind“ oder „Ozempic-Babies“.

Ozempic ist der Handelsname für den GLP-1-Rezeptoragonisten Semaglutid als Diabetespräparat, die höher dosierte Variante zur Gewichtsabnahme mit demselben Wirkstoff heißt Wegovy – beides vom Hersteller Novo Nordisk.

Im Wesentlichen werden für die erhöhte Wahrscheinlichkeit, schwanger werden zu können, 2 Gründe diskutiert: Zum einen könnte die Gewichtsabnahme als solche dafür verantwortlich sein, zum anderen eine Veränderung der Magen-Darm-Passage und konsekutiv einer Veränderung der Resorption von Hormonpillen, die zur Verhütung eingenommen werden.

- Ozempic-Babies (Wegovy)
- possible explanations
 - weight loss and normalization of hormonal balance
 - altered stomach passage

Example absorption models

- A. first-order absorption with different rate of absorption
- B. first-order absorption with different lag-times
- C. Transit chain ($n=3$, different rates)
- D. Transit chain (different n)

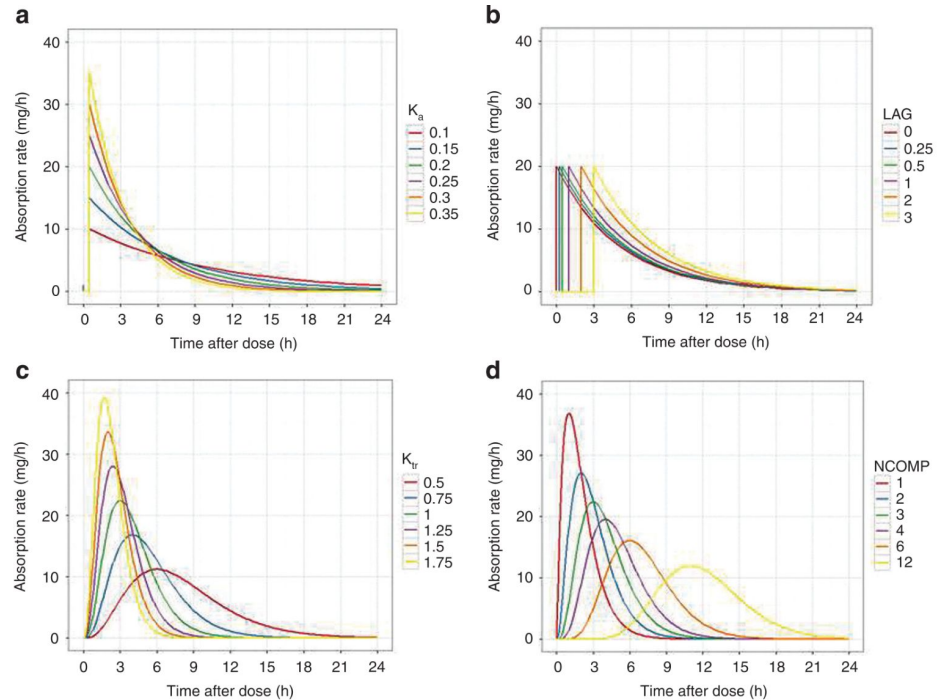


Figure 1 Models of extravascular absorption. The time profile of absorption rate for selected absorption models. **(a)** A first-order absorption model with different values of the absorption rate constant (k_a). Absorption lag was 0.5 h in all cases. **(b)** A first-order absorption model with different values of the absorption lag (LAG). Absorption rate constant was 0.5/h in all cases. **(c)** A three-compartment transit chain model with different values of the transit chain rate constant (k_{tr}). Note that decreasing the rate constant lowers the overall absorption rate and delays the time of its maximum value. **(d)** Transit chain models with different numbers of transit chain compartments (NCOMP). The transit chain rate constant was 1/h in all cases. Note that increasing the number of compartments introduces a delay before absorption, and functionally acts as a lag. The dose was 100 mg in all cases (hence the area under the curve should be 100 mg for all models).

Mould DR, Upton RN. **Basic concepts in population modeling, simulation, and model-based drug development-part 2: introduction to pharmacokinetic modeling methods.** CPT Pharmacometrics Syst Pharmacol. 2013 Apr 17;2(4):e38. doi: 10.1038/psp.2013.14. PMID: 23887688; PMCID: PMC3636497.

Drug formulation



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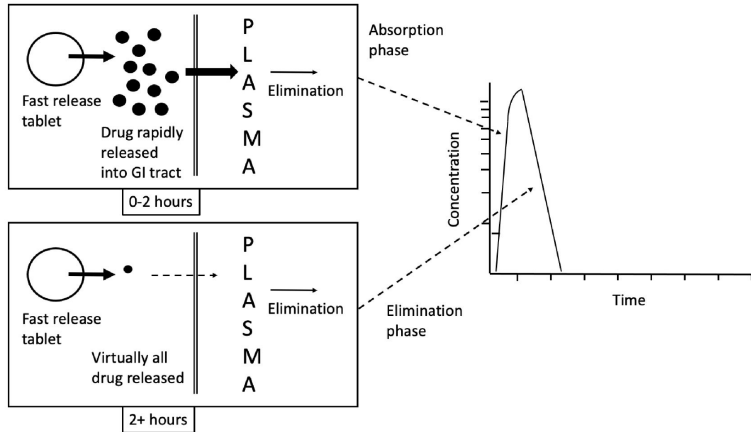


Figure 8: illustration of how the absorption and elimination phases are affected by rapid release of drug into the gastrointestinal tract (GI Tract).

t Lappin & Seymour, 2022

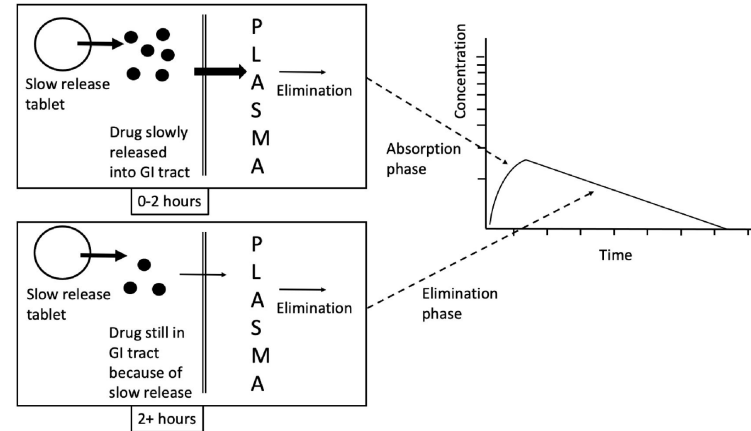


Figure 9: illustration of flip-flop kinetics caused by slow release of drug into the gastrointestinal tract (GI Tract).

Variability in Dose & Dissolution

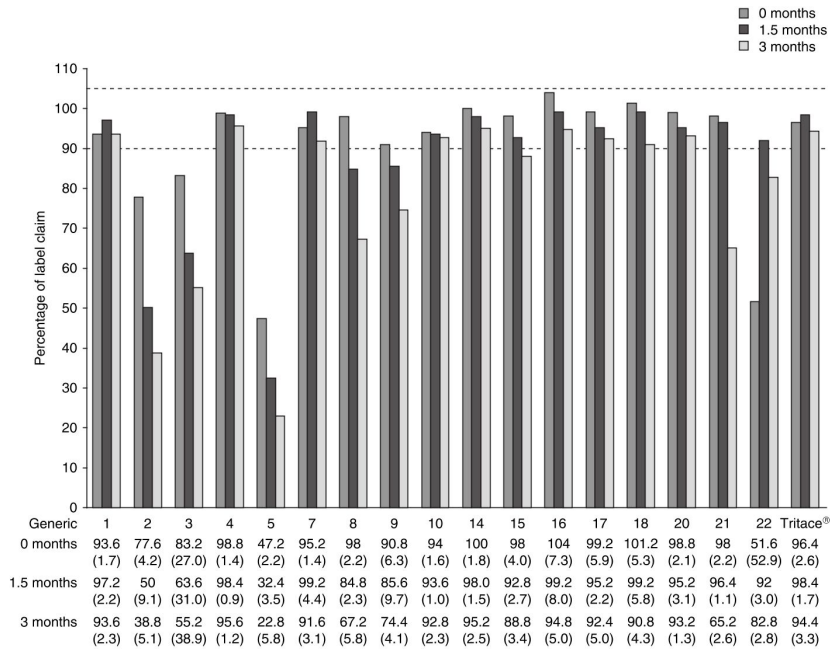


Fig. 2. Assay of ramipril levels vs label claim. Percentage of ramipril in 17 ramipril generics/copies vs the reference ramipril product (Tritace®) before and after 1.5 and 3 months of temperature-stressed storage, as determined using a validated high-performance liquid chromatography system (see Materials and Methods section for details of the assay).^[16] Results represent the mean (relative standard deviation) of the average values for ten individual tablets. No data are reported for Generics 6, 11, 12, 13 and 19 as the assays for these were not completed because of a shortage of tablets. Sanofi-Aventis specification limits (within 90–105% of label claim) are illustrated by the dotted lines.

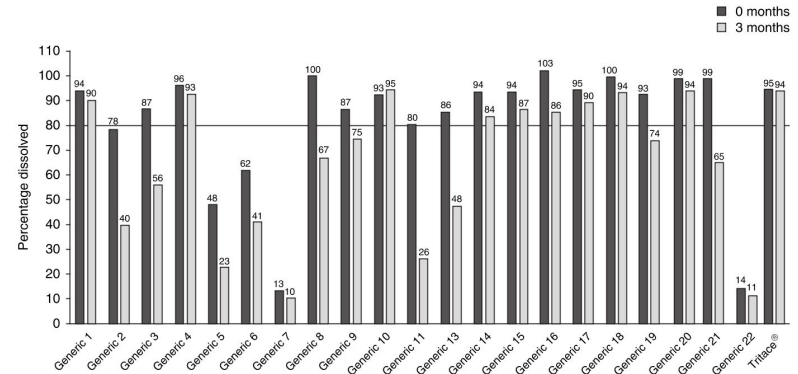


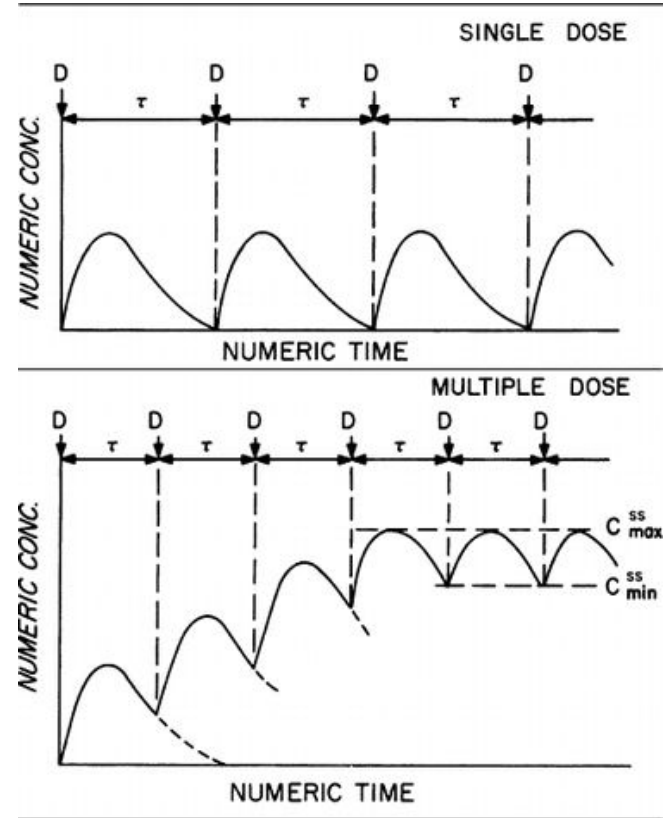
Fig. 4. Dissolution profiles of 21 ramipril generics/copies vs the reference ramipril product (Tritace®). Results from 30-minute dissolution assays conducted at baseline (day 0) and after 3 months' temperature-stressed storage; results are the mean of the average values from six individual tablets. The Sanofi-Aventis specification limit ($\geq 80\%$ dissolution in 30 minutes) is illustrated by the black line. No data are reported for Generic 12 as the dissolution assay was not completed because of a shortage of tablets. Dissolution assays were conducted according to validated dissolution tests for solid dosage forms using a paddle system (Apparatus 2 as described in the European Pharmacopoeia 6.0^[17]); dissolved contents were assayed using a validated high-performance liquid chromatography system (see Materials and Methods section for details of the assay).^[16]

Angeli DG, Trezza C. **Quality and stability of ramipril generics/copies versus reference ramipril (Tritace): a 3-month stability comparative study.** Clin Drug Investig. 2009;29(10):667-76. doi: 10.2165/11315270-000000000-00000. PMID: 19715383.

Multiple Dosing

Multiple dosing in drug therapy refers to the practice of administering multiple doses of a drug or medication

1. **Steady-state Concentration:** Over time, multiple doses of a drug lead to a steady-state concentration in the blood.
2. **Dosing Interval:** The dosing interval is the time between two consecutive doses.
3. **Therapeutic Window:** This is the range of drug concentrations in the blood that provide effective treatment without causing toxicity.
4. **Dose and Frequency:** The size of the dose and frequency of dosing are critical in multiple dosing strategies.
5. **Accumulation and Elimination:** When drugs are administered repeatedly, they can accumulate in the body, leading to higher concentrations than after the first dose.
6. **Interindividual Variability:** There can be a significant difference in how different individuals respond to the same drug and dosage due to genetic factors, age, gender, organ function, and the presence of other diseases or drugs.
7. **Compliance:** Compliance with the dosing regimen is essential for the success of the treatment.

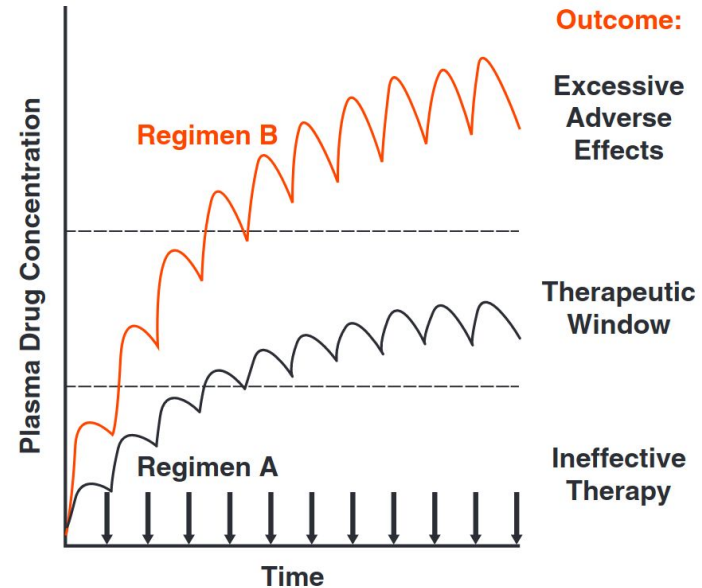


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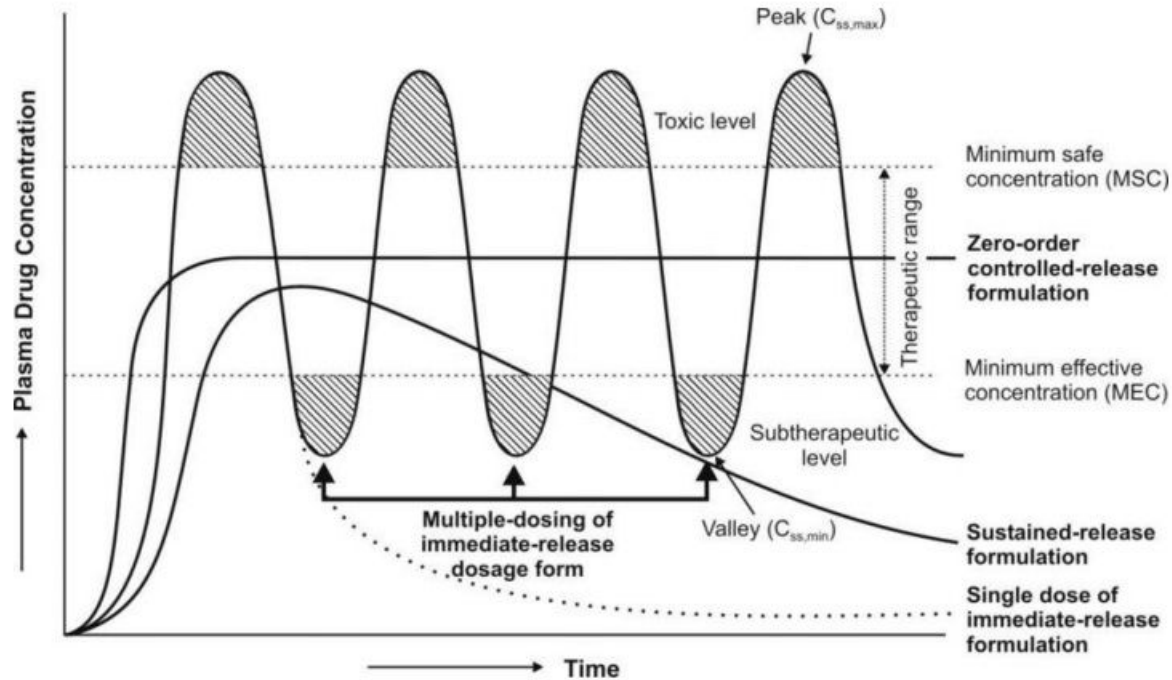
Optimal Dosing Regime & Therapeutic Window

The therapeutic window is determined by two main factors: **the minimum effective concentration (MEC)** and the **maximum tolerable concentration (MTC)**.

- The **minimum effective concentration (MEC)** is the lowest concentration of a drug in the patient's bloodstream that still produces the desired therapeutic effect. If the concentration falls below this level, the drug may not be effective in treating the condition, leading to ineffective therapy.
- The **maximum tolerable concentration (MTC)** is the highest concentration of a drug that can be tolerated without causing significant toxic effects or side effects. If the drug concentration exceeds this level, the risk of side effects and toxicity increases.



Controlled release medication



Compliance

- after the end of first year only ~50% continue to take the prescribed medication
- anti-hypertensive drugs

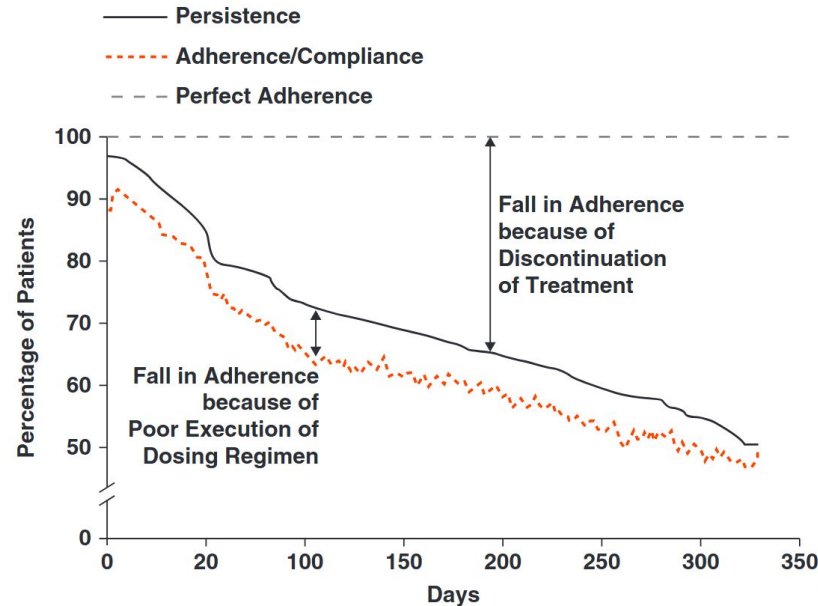


FIGURE 1-9 Nonadherence to prescribed medication is a major source of variability in drug therapy. Shown is the gradual but persistent decrease in adherence in the percentage of 4783 patients prescribed a variety of once-a-day antihypertensive therapies due to discontinuation of treatment, such that by the end of the first year only 50% of the patients prescribed the treatment for an indefinite duration continue to take the prescribed medication. The initial 3% drop in adherence is due to some patients never even starting the medication. The data were obtained using an electronic monitoring device that detects and logs each time the container with the medication is opened. (Taken from Vrijens B, Vincze G, Kistanto P, et al. Adherence to prescribed antihypertensive drug treatments: longitudinal study of electronically compiled dosing histories. *Brit Med J* 2008;18:1–6.)